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RESEARCH ARTICLE

Enhancing Web Development Education With Game-Based and Gamification Learning: A Study of Engagement, Motivation, and Performance

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ABSTRACT Web development is a dynamic and complex subject in the computer science field that can be challenging to learn among higher education students. We investigated the impact of game-based learning and gamification on students' engagement, motivation, and performance in a web development course in this study. We tested 60 university students, who were expected to learn from a novice or beginner level. The experimental group used a gamified system, while the control group used a non-gamified system. All game elements in this study were grounded in self-determination theory (SDT), which suggests that intrinsic motivation increases when learners feel autonomous, competent, and connected, and flow theory, which emphasizes that when challenge and skill are balanced, learners are more likely to experience a state of flow. Quantitative and qualitative data were both collected and analyzed. Results showed that the gamified system positively impacted students' performance, compared to the non-gamified system. Interestingly, there was no significant difference in the engagement and motivation of the students. The findings of this study suggest that while the game elements addressed students' psychological needs, gamified learning outcomes depend not only on the inclusion of these game elements but also on individual learner differences such as prior game experience, learners' expectations of the gamified system and how well the designed system sustains engagement over time to overcome the novelty effect. In conclusion, these findings have significant implications for web development education and suggest that game-based learning and gamification are promising approaches for web development education.

INDEX TERMS Game-based learning, gamification, web development, teaching and learning strategies.

I. INTRODUCTION

Web development is a fundamental component of computer science, and it involves learning web concepts, web technologies, markup languages like HTML, style sheet languages like CSS, and programming languages. The demand for web development professionals is projected to remain strong through 2033, with the growth in software development, cybersecurity, and AI technologies. According to

recent projections, computer and mathematical occupations are expected to grow by 12.9%, with software developers expected to see a 17.9% and web developers a 9.0% increase in employment, both faster than the average for all occupations. In fact, with the growing advancement in AI technologies and the modernization of businesses, development of these emerging technologies will require professionals in computer science, programming and software development. As a result, the demand for web development skills is expected to rise in response to emerging technologies [1], [2]. With this significant growth, the need to

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integrate web development in higher education continues to increase [3]. Besides, there is increasing attention to improving students' performance in web development, so that they can be capable of solving real-world problems [4]. However, teaching and learning web development in higher education remains challenging.

Prior to this study, we conducted a baseline literature review to understand the challenges of learning web development and it revealed that in learning web development, students are often confronted with the inherent complexities of web development concepts [5], [6]. This difficult nature of web development can overwhelm students and lead to frustration that negatively impacts their engagement and motivation. Also, the conventional instructional methods used may not effectively address these complexities, as they are also noted to contribute to how well students can comprehend programming concepts [7]. Previous studies show that the absence of interactive learning compounds the challenges of learning such subjects [8], limits students' hands-on practice opportunities, contributes to their lack of motivation, and ultimately affects their skill development [5], [9]. The result is that students are unable to develop applications or bridge the gap between theoretical knowledge and real-world application [10], [11], [12]. In studies that have investigated the pedagogical demands of teaching and learning STEM subjects like web development, one recommendation that arises is the integration of educational technologies that have proven to be effective [13], [14], [15], [16], [17]. These technologies, including online interactive learning, digital gaming, simulations, augmented reality, virtual reality, and artificial intelligence serve as effective tools for engaging students in thinking through complex concepts [18], [19], [20].

Previous studies show that the integration of GBL and gamification approaches have shown promise in various educational contexts [21], [22], [23], [24], [25], [26]. These approaches have increased students' engagement, access to lessons, and enhanced their academic achievements [27], [28], [29], [30]. Despite the considerable speculation about the positive effect of games, this method is not a common approach in higher education for learning web development, and empirical research on the effectiveness of gamification is limited [31]. However, in broader education settings, it is revealed that games are now very often used in classrooms [32], with about 86.8% of teachers already utilizing games to enhance their lessons and 18.8% using them daily [33]. Despite this increasing usage, other studies suggest limited implementation of this approach, with one study finding only 30% of teachers using gamified tools [34]. Overall, teachers are noted to have a positive attitude towards the use of gamification [35], as this has been recognized to increase engagement and motivation [36]. On the other hand, some educators have expressed a potential negative effect it can have when used too much [37]. These findings suggest that the integration of games in education has shown mixed results. Moreover, there are limited empirical studies with

many suffering from methodological problems such as a lack of comparison groups, short treatment, singular assessments, and lack of validated measures [38], [39], [40].

Considering the challenges faced when learning web development, three of the holistic elements that can be influenced are student engagement, motivation, and performance. These variables are necessary for an effective and successful learning experience [41], [42], and they have been extensively researched and are widely recognized as key drivers of educational outcomes [41]. Therefore, with these views in mind, this study intends to explore the effect of Game-based learning (GBL) and gamification learning approaches on student engagement, motivation, and performance within the specific context of web development, utilizing a gamified web-based system that incorporates GBL and gamification and its comparative analysis with a non-gamified web-based system, which does not incorporate GBL and gamification.

To guide the design and interpretation of the gamified learning experience, this study is guided by two key psychological frameworks, Self-Determination Theory (SDT) and Flow Theory, which explain how specific gamified elements can support students' engagement, motivation, and learning outcomes.

Three research questions (RQs) were formulated informed by the literature findings: (RQ1) Does the use of the gamified system in learning web development lead to higher engagement of students in the gamified group (experimental group) in contrast with the non-gamified group (control group)? (RQ2) Does the use of gamified in learning web development increase students' motivation to learn web development compared to the non-gamified system? (RQ3) Do the students who used the gamified system have higher performance than students who used the non-gamified system?

The rest of the paper is organized as follows:

The Related Work section examines previous studies on game-based learning and gamification in STEM education and discusses the research gap this study aims to address. The Building on SDT and Flow Theory, the Theoretical Framework section describes the psychological mechanisms that connect gamification to motivation and engagement. The Methodology section describes the participants, web-based system design, data collection, and analysis. The Results section presents the result of the study, focusing on students' engagement, motivation, and performance. Finally, we concluded the paper with a Discussion and Conclusion section with a summary of our findings, practical implications and suggestions for future research direction.

II. RELATED WORK

A. WHAT IS GAME-BASED LEARNING AND GAMIFICATION

GBL and gamification share common objectives, but they differ in implementation. GBL incorporates games into the learning process [43], [44]. On the other hand, gamification [45] takes gameplay elements and applies them to the existing learning courses to motivate and

engage learners [44]. These interactive learning approaches cater to modern learners who often respond positively to technology-enhanced and game-like learning environments [41], [46]. Their effectiveness is increasingly being understood through psychological frameworks such as SDT and flow theory, which explains how game elements, like levels, points, badges, and leaderboards can support learners' engagement, motivation, and provide sustained learning experiences.

B. ENGAGEMENT, MOTIVATION, AND PERFORMANCE IN WEB DEVELOPMENT

The idea of Engagement is to ensure that students are actively involved in learning, and motivation promotes their desire to learn a subject, and performance on the other hand reflects students' capability to apply knowledge, thus, creating a learning experience that can set them up for success in this subject [41], [42], [47]. The role of engagement, motivation, and performance have been seen as important factors for successful educational outcomes. In previous studies, enhancing engagement led to improved learning outcomes for students and contributed to their professional development [48], [49]. As we can see in other reports, motivated learners tend to perform better academically [50].

We can imply that the significance of these variables aligns with the overall learning experience in web development, as evidenced by [51] research, which highlighted the enhancement of students' engagement through active learning in web development courses. In the study, the unique approach of integrating quizzes was designed as a means of assessment as well as mechanisms to trigger reflection and metacognitive activities, to enhance both engagement and motivation throughout the learning process, which in turn would improve course outcomes. Student performance plays a critical role in shaping their learning outcomes and ability to excel. For example, in [52] and [53], these studies show how interactive learning media on web programming results in better learning results, engagement, and motivation. These studies offer valuable insights into engagement, motivation, and performance, and how learning supported by engaging and adaptive tools can enhance both learning outcomes and student success in web development education. Building on this, our study also considers how these outcomes relate to key psychological factors that inform the design of gamified learning systems.

C. GAME-BASED LEARNING AND GAMIFICATION IMPACT ON STUDENTS' ENGAGEMENT, MOTIVATION AND PERFORMANCE

According to studies, digital games have been used to support the engagement, motivation, and performance of students in a range of academic settings to improve their learning experiences [28]. In several studies [21], [22], [23], [24], [25], [26], GBL and gamification approaches have shown promise in different educational contexts, including increased access to lessons for students, improved engagement, and

enhanced academic achievements [27], [29], [30]. Game elements are being used more in education to get students more engaged. Recent work has highlighted the combined impact of social learning and gamification on student engagement and academic achievement in higher education. For instance, [54] conducted a longitudinal study using a gamified Moodle platform and found strong correlations between the participation patterns of gamified activities with knowledge acquisition and final learning outcomes.

'KAHOOT!' is another good example. It shows how game-based platforms can make traditional learning better. In a study done with AP Biology students, using 'KAHOOT!' showed a high level of engagement, and improved understanding of difficult topics like transcription and translation [55]. Another study assessed the emotional engagement of adult participants using game-based and non-game-based learning methods. They showed the power of game elements, particularly when coupled with effective feedback mechanisms [56]. The application of games in an educational context is seen as a powerful strategy to keep students engaged [28], [57]. These findings highlight the unique benefits of using GBL and gamification in educational contexts. Despite this impact of game-based learning and gamification on engagement, other studies have also noted that gamification might also fail to keep users engaged and can also vary depending on the different aims of the systems being gamified [57], [58], [59]. These varied findings also suggest that the success of gamified learning may depend on how well basic psychological needs that are known to influence learners' engagement and motivation are supported in the design of the gamified experience [60].

Learners' motivation is the driving force behind their willingness to engage in the learning process. Reference [61] used this strategy and found that incorporating game elements into a course increased student motivation, resulting in higher motivation and achievement in the experimental group compared to the control group. The study in [62] after using gamification, noticed a big difference in motivation scores between the groups that tried it and those that didn't. Accordingly, other studies [63], [64] have also indicated that interaction with gamified elements in a learning context can increase learners' motivation, although further application of these findings is necessary to confirm the effectiveness of this approach [64]. Gamification has also been reported to cause boredom among students leading to a lack of motivation to continue learning, and game elements like challenges, and leaderboards, can lead to problems in maintaining the engagement and motivation of users [37], [65], [66].

Parallel to these studies, [67] also examined the link between these approaches and academic performance. They reported that the experiment group, which was exposed to GBL strategies, showed significantly higher post-test scores compared to the control group receiving traditional instruction. Similarly, a systematic literature review and meta-analysis examining 26 empirical studies supported these results, showing that GBL, especially when grounded in

constructivist theories, consistently led to knowledge attainment and improved academic outcomes [68]. Also, it is seen that the introduction of gamification in computer science courses led to noticeable improvements in students' grades over time [69]. Similar trends were observed, with students participating in gamified systems outperforming their counterparts in non-gamified settings [69], [70]. Despite the noteworthy impact of games in improving students' performance, other studies [65], [66] have also noted a lack of improvement in students' performance that used gamified systems.

Moreover, while the traditional applications of game-based learning and gamification have demonstrated significant impact on students' engagement, motivation and performance, recent advancement in technology have extended these findings to AI-enhance gamification, immersive educational gamification and adaptive learning environment across diverse educational settings including higher education, secondary and primary schools, in specialized fields such as language learning and medical fields. For example, a recent study [71] examined the effects of gamification and automated questions generation (AQG) on student's engagement and performance using a Moodle plugin to deliver interactive, personalized assessment. The findings demonstrated a significant boost in student motivation and learning outcomes compared to traditional methods. Similarly, [72] and [73] found that AI-powered gamification can significantly improve medical students' learning outcomes and strengthen their practical and cognitive skills. These studies highlight the potential of AI to personalize and enhance gamification education. In addition, research into the role of immersive educational gamification in strengthening learning experiences has also been highlighted to improve learners' motivation, engagement and academic achievement. Particularly, a systematic review by [74] reviewed seventeen studies and found that integration of game-based learning and gamification technologies such as immersive reality, learning management systems (LMS) and massive open online courses (MOOC) have shown promise in engineering and healthcare fields, despite challenges such as accessibility issues, teachers' competency and insufficient research linking learners' use of these strategies to learning outcomes. Based on its proven effectiveness in diverse settings, applying this approach to web development shows great promise.

D. GAME ELEMENTS AND MECHANISMS, AND THEIR IMPACTS

Progress tracking, reward mechanisms, and goal-focused activity are said to be the three basic parts of most games [75]. Gamification in education shares these aspects and includes points, challenges, badges, and more [76]. The major reason behind the application of gamification in education as highlighted by [77], is because it emphasizes achieving specific goals, using diverse reward systems like leaderboards

and achievements, and tracking progress. The common focus on goals, rewards, and progress creates effective gamification, not only in education but also in motivating, engaging, and achieving goals in broader contexts. Game mechanics are crucial elements within game applications. They are often used to make learners more engaged and motivated [77], [78]. Considering their influence on engagement and motivation, it is important to thoughtfully consider and apply them, as their careful implementation can profoundly impact the overall gamified experience [26], [78]. Reference [79] identified common game mechanics, which include:

- 1) Challenges, Trophies, Badges, Achievements.
- 2) Points.
- 3) Levels.
- 4) Virtual Goods.
- 5) Leaderboards.
- 6) Competitions.

These game mechanics can be strategically applied to engage and motivate users in various contexts. Whether utilized individually or in combination, they have the potential to stimulate motivational aspects among users especially when aligned with learners' psychological needs [79]. The game mechanics utilized in this study include levels, points, badges, challenges, and leaderboards. The decision to include these specific game mechanics was informed by careful consideration of their positive impact, as well as insights and findings identified from previous studies that supported their ability to enhance user engagement, motivation, and learning performance.

Previous study has highlighted that these game features are important for keeping players interested and motivated. For example, badges have been recognized as significant contributors to sustaining engagement [80] and improving learners' performance [81]. When incorporated with social learning platforms, by leveraging surprise rewards, game points and activity points as a means of leveling up and earning these badges, they will not only positively influence learners but also provide earlier indicators of academic success, significantly enhancing engagement, motivation and performance [82]. It has also been emphasized that the presence of badges showed no significant improvement in learners' motivation or performance [80], [83].

Similarly, points and leaderboards have been identified as contributing factors to engagement and effective motivators for achievement [84], [85]. Levels serve as indicators of reaching milestones. They provide learners with a sense of achievement and have been observed to be utilized in conjunction with other gamification elements and have been noted to positively impact on student motivation and engagement [86]. Challenges were used to progressively build learners' sense of competence and support autonomy by allowing self-paced progress [60]. To ensure that our study is well-supported, we base our choices on existing research

to provide a strong theoretical foundation for applying these game mechanics.

E. EMPLOYING WEB-BASED GAMIFICATION SYSTEM

Web-based systems like learning management systems (LMS) [87] are well-suited to meet the expectations of digital-native students. The use of gamification in a web-based system allows the application of these game mechanics in a web-based learning environment such as LMS that manages the learning process of students. LMSs are said to be suitable for gamification because they can automatically track students' progress and results. They also allow administrators to create levels, points, and badges that learners can earn for completing activities [88]. This creates a sense of interactive learning.

Wordwall is another web-based interactive environment to create activities and games to enrich students' learning process. It is said to offer students an active learning experience by engaging with content at their own pace [89], and its interactive games have been used as a learning media in improving students' learning outcomes [90], and as a review for materials learned. Web-based gamification systems have been particularly seen to be effective in improving learners' creative skills, illustrating complex concepts, and improving their problem-solving skills [91], [92], [93]. These platforms' capabilities support our focus on enhancing engagement, motivation and performance of students in web development education. By combining these two web-based tools that support learners' active interaction with course materials, receiving immediate feedback, and progress tracking, we ensured that the technological environment complemented the psychological goals as further discussed in our theoretical framework.

F. GAMIFICATION OF WEB-BASED SYSTEM FOR INSTRUCTIONAL DESIGN

When applying gamification to an educational context, such as in web-based systems, to increase students' engagement and improve learning, the lesson plan can be developed by applying a set of steps process [57], [94], which continues to influence recent implementation of gamified learning systems [95]:

- 1) Understanding the target audience or learners' needs.
- 2) Defining the objectives of the lesson.
- 3) Structuring the content and Activities by considering credible information that can be used to inform the implementation of the instruction that supports the progress of the students. Structuring the content in stages or milestones can make the overall objective seem more achievable for students.
- 4) Evaluating and choosing the resources, by considering the resources needed to gamify the education.
- 5) Applying gamification elements, like points, levels, leaderboards, badges, and challenges to lead learners to achieve their objectives.

G. PRESENT STUDY SIGNIFICANCE AND RESEARCH HYPOTHESES

While prior studies have explored GBL and gamification in STEM education, much of the empirical literature focuses on general programming courses, mathematics, or science courses. However, the effects of game-based and gamification learning in web development education remain underrepresented in the existing empirical literature.

Our study addresses this gap by exploring the impact of a web-based gamified learning system in the particular setting of web development, where game-based learning and gamification approaches are often lacking. Unlike previous studies, the present study integrates both practical and theoretical elements grounded in established psychological frameworks into the web development course.

Moreover, this study offers a quasi-experiment evaluation of the effects of these approaches on students' engagement, motivation, and performance. By aligning the web-based gamified platform with specific course objectives and motivational design principles, this study provides new empirical evidence that extends prior studies and provides insights for educators in web development education. Based on this, we hypothesize that using the gamified system will have a significant impact on students' engagement, motivation, and performance.

III. THEORETICAL FRAMEWORK

The use of gamified approaches in this study is guided by their increasing evidence and ability to improve learners' engagement, motivation, and academic performance. Prior studies have also shown that game elements such as levels, badges, points, challenges, and leaderboards can improve learners' experience in different academic settings. To provide a strong theoretical foundation for our design choices, this study draws on two theories (Self-Determination Theory (SDT) and Flow Theory) that have been used to explain how game mechanics influence learners' engagement and motivation.

A. SELF-DETERMINATION THEORY

SDT is a key framework that helps understand the psychological mechanism underlying learners' motivation [96]. This theory accounts for a full spectrum of motivation from extrinsic (driven by external rewards) to intrinsic (driven by internal interest and enjoyment) and has been widely applied in several studies to explore the psychological processes that drive why learners feel motivated in gamified environments [97]. SDT suggests that extrinsic motivators can become internalized and take on qualities that are more intrinsic when the learning environment supports three basic psychological needs, which are autonomy (a feeling of being in control of one's action), competence (a feeling of being capable and mastery in a domain) and relatedness (a feeling of being connected to others) [96], [98]. When these needs are met, learners are likely to experience intrinsic motivation, which has been

linked to engagement, better performance, and sustained learning [99], [100], and gamified learning adeptly addresses these needs [101], [102].

In our study, these psychological needs are met through the design of the game mechanics. For example, the quest-based challenges in the gamified system were used to give students a sense of autonomy, by allowing the students to complete tasks and progress at their pace. The progress system (levels, points, and badges) provided a sense of competence where student can see their improvement, feel capable, and experience mastery of the course. Additionally, the leaderboard introduced a feeling of relatedness by making students have a sense of connection with others. Based on SDT, we designed the gamified experience to support learners' intrinsic motivation by addressing their needs for autonomy, competence, and relatedness.

B. FLOW THEORY

Flow theory, developed by Csikszentmihalyi [103], suggests a perspective for understanding engagement by highlighting the importance of immersive learning experiences. A student who is in a flow state feels deeply focused, in control, and experiences satisfaction when performing an activity [104]. According to this theory, when challenges and skill are balanced, learners are more likely to experience a state of flow [105]. Games support this by creating challenging, goal-oriented activities that are neither too easy nor too difficult but are continuously adapted to the player's skill level [106]. Also, immediate and meaningful feedback is key to maintaining this type of flow [107].

According to previous studies, the use of progress bars, badges, levels, and leaderboards often supports flow by helping students progress, stay focused and engaged through an activity [108], [109], [110]. In our study, we used quest challenges with increasing difficulty to maintain the challenge and skill balance and keep students in a state of flow as their knowledge progressed through the web development course. To provide the students with clear goals and immediate feedback, we used progress points, badges and levels, which also reinforces their sense of competence. Additionally, the use of the leaderboards added a layer of relatedness that can further contribute to increased participation and enhance the students' engagement. These theoretical frameworks were fundamental for helping us understand the effectiveness of the game mechanics we've used and how they influence the student's motivation and engagement.

SDT guided our selection of mechanics that support intrinsic motivation, and flow theory informed how we can structure the learning experience to support students' progress and immersion by maintaining a balance between the challenge and their skill. We used quests, progression system, and social feature with specific psychological goals that support learners' autonomy, focus and mastery. Our gamified system was created with a clear psychological goal by integrating SDT and flow theory. Its goal was to create

a meaningful learning experience for students, not just to encourage them with rewards. The gamified elements were designed to support the students' engagement and motivations, as well as contributing to their learning outcomes.

IV. METHODOLOGY

A quasi-experimental design with non-equivalent groups was used, with a mixture of quantitative and qualitative. The participants were divided into two comparison groups: an experimental group (gamified group) who received the gamified web-based system and a control group (non-gamified group) who received the non-gamified web-based system with the same course content, each consisting of 30 students. The study received approval from school authorities, and all students provided written informed consent. The ethical committees also approved all research processes and data collection methods.

A. PARTICIPANTS

This study included 60 (aged 18 – 32 years) university engineering students in the experiment, selected from two universities located in Turkey in the fall semester. Random assignment of participants to experimental and control groups was not possible due to logistical constraints, however, participants were carefully grouped to ensure that they similar knowledge levels, as well as comparable characteristics in terms of, age, and level of education. As an inclusion criterion, our study included individuals with undergraduate or master's degrees to maintain focus on learners in the earlier stages of their academic or professional journey in web development. By doing this, our study aimed to control for potential confounding variables that could have influenced the results and increase the internal validity of the study, ensuring that the groups were equivalent before the intervention. There were 30 students that enrolled for the experiment for each group. However, 1 participant was removed from the control group and 6 from the experimental group based on the inclusion criteria. Hence, a total of 53 students were considered for this study. We analyzed the distribution of the groups (experimental and control groups) based on age, gender, level of education, pre-knowledge to help us understand the participants' knowledge level in web development, and pre-test scores to ensure equal distribution.

1) PARTICIPANTS' EDUCATIONAL LEVEL, GENDER, KNOWLEDGE LEVEL AND PRIOR EXPERIENCE

In this study, 75% of the control group and 83% of the experimental group were bachelor (undergraduate) students, aligning with focus on learners in the earlier stages of their academic or professional journey in web development. The gender distribution for experimental group (Male = 71%, Female = 29%), while control group (Male = 62%, Female = 38%). The percentage of males in both groups align with the typical gender distribution of STEM students in this country. Additionally, the gender distribution within each group can be considered similar for the purposes of this study.

The Shapiro-wilk test with $p = 0.170$ and the Levene's Test for Equality of Variances with $F = .826$, $p = 0.368$, indicating assumptions are met. An independent t-test was conducted, and the test was not significant, $t(51) = 1.55$, $p = 0.126$, with mean average for experimental group ($M = 46.33$, $SD = 18.05$) and control group ($M = 53.72$, $SD = 16.54$), indicating that the two groups prior level of web development knowledge were comparatively equal.

For the prior experience (question sample "Have you ever tried learning web development?"), approximately 63% of the experimental group and 57% of the control group reported not having tried learning web development before. The proficiency survey (question sample "I know web development very well") responses show that the majority from the two groups (experimental group and control group = 72%) either disagree or strongly disagree. Based on these results, we assume that the participants had limited prior knowledge and familiarity with web development concepts. Therefore, participants were expected to learn web development from a novice or beginner level.

This novice or beginner level was especially appropriate for our study's objective to assess how gamified learning can support learners in building fundamental skills in web development. Gamified systems that are designed to support engagement, motivation, and performance are particularly helpful for learners at this stage developing fundamental skills [111], as they tend to respond well to a learning environment that provides feedback and motivation support [76], both of which are elements that align with the theoretical foundations of this study.

B. DEVELOPING OF THE LEARNING AND GAMIFIED SYSTEM

The development process of the learning system followed the industry's best practices, including researching and analyzing relevant documents. This helped create a web-based system that integrates learning management and Wordwall tools for delivering the course and game elements [112], [113]. To design the web-based systems effectively and incorporate game elements, we went through the following process [57]:

1) UNDERSTANDING THE STUDENTS' NEEDS

Previous studies have shown that students find web development concept complex. The conventional instructional methods employed might not effectively address these challenges, as they are also noted to contribute to how well students can comprehend web technologies or concepts. Additionally, the lack of interactive learning further complicates learning such subjects. To address these challenges, we have developed a gamified web-based system with an introductory topic on HTML. The course covered topics which included introductory topics on HTML.

2) DEFINING THE LEARNING OBJECTIVES

A web development subject HTML was provided to students in the fall semester. The course covered introductory topics such as HTML anatomy, headings, paragraphs, line breaks, and HTML attributes. Bloom's taxonomy [114] which was later revised was used in developing the learning objective of this course [115]. For this course, three lessons were covered, and with the application of Bloom's taxonomy, the lesson objectives were created for each lesson (See Table 1). The course had eight objectives and Bloom's taxonomy was applied to help students build knowledge based on previous lessons.

Bloom's taxonomy was used to create a table of specification (TOS) to guide test construction and ensure content validity (See Appendix A Table 8). Two domain experts from computer engineering teaching and education technology teaching also helped to better understand the requirement of the course development and materials.

TABLE 1. Lesson Objectives.

Lessons	Lesson Title	Learning Objectives
Lesson 1	Introduction to HTML	Objective 1: To demonstrate the ability to define what HTML stands for. Objective 2: To demonstrate the ability to identify simple HTML document components.
Lesson 2	HTML Anatomy	Objective 3: To recognize how an HTML element is written. Objective 4: To be able to identify what a body element is used for. Objective 5: To be able to determine the difference between parent and child elements.
Lesson 3	Headings, Paragraphs, Line Break, and Attributes	Objective 6: To demonstrate the ability to apply HTML tags like heading, paragraphs, and line breaks tags correctly. Objective 7: To understand what HTML attributes are. Objective 8: To determine the correct use of attributes.

3) DESIGNING THE NON-GAMIFIED AND GAMIFIED

The non-gamified system was developed without any integrating of game elements and focused solely on presenting instructional content in a traditional web-based manner (See Appendix C Figure 1).

The gamified system incorporated games elements to enhance students' engagement, motivation, and performance. The web-based interface included interactive components to create a more enjoyable and progressive learning experience (see Appendix C Figure 2). The combination of

LMS, Wordwall, and WordPress content management system (CMS) was used as integral tools in designing and developing the gamified system. Based on the learning objectives and lessons, three game plays were embedded in the gamified system. The first gameplay is a maze chase game that includes 4 questions to help students get familiar with the concept of HTML according to objectives 1 and 2. Students can play the game after completing lesson 1 (See Appendix C Figure 3). The second game play is a maze chase game that includes 5 questions to help students understand HTML's anatomy according to objectives 3, 4 and 5. Students can play the game after completing lesson 2 (See Appendix C Figure 3). The third gameplay is a maze chase game that includes 10 questions to help students to demonstrate the ability to apply HTML tags and the correct usage of attributes according to objectives 6,7 and 8. Students can play the game after completing lesson 3 (See Appendix C Figure 3). Table 2 shows the game elements that were applied, and the variables that we presumed they would enhance as reviewed from previous literature.

4) DESIGNING THE EXPERIMENT

The experiment consisted of two subject groups, control and experimental group. The study was designed with three stages as illustrated in Figure 5. In the first stage, both groups were given a demographic questionnaire, a motivation prequestionnaire, and a pre-test.

In the second stage, after completing the pre-test, each group was redirected to the non-gamified and gamified web-based system to complete the activities like taking the course lessons alongside the gameplays. In the third stage, each group was redirected to take post-engagement questionnaire, a post-motivation questionnaire, and a post-test to complete the experiment.

C. DATA COLLECTION AND ANALYSIS

Data was collected using tests and questionnaires, allowing for both quantitative and qualitative data collection methods.

The questionnaire assessed the students' engagement, adapted from existing instruments [116], [117], [118], [119]. The measured items of this part are five-point Likert scale statements. All the scale points of the items were labelled, ranging from 1 ("Strongly disagree") to 5 ("Strongly agree") (sample items: "The activities in the system were engaging").

To minimize bias [120] the measured items were adapted from multiple sources. Adapting existing developed instruments [118], [121] the questionnaire assessed the students' motivation.

The measured items of this part are five-point Likert scale statements. All the scale points of the items were labelled, ranging from 1 ("Strongly disagree") to 5 ("Strongly agree") (sample items: "I felt confident using the system"; "Using the system I felt in control of my learning process"). Finally,

TABLE 2. Game elements application in web-based gamified system.

Game elements	Application description	Presumed impact
Progress points	Students earn progress points when they finish specific tasks, such as completing lessons and playing game activities (see Appendix C Figure 4). Accumulated points were associated with level increase, providing positive reinforcement for active participation. The maximum progress points a student can earn is 100 points, showing that they have completed the entire set of lessons within the system, serving as a measure of achievement in learning the introductory HTML concepts [76].	Engagement [84]
Levels	A tiered leveling system was implemented, allowing students to progress through different levels (see Appendix C Figure 4). There are three levels, as students keep progressing in the lessons, and winning more points, their level keeps increasing. Each level introduced more advanced HTML concepts, providing a structure learning path [86].	Motivation, Engagement [86]
Badges	Virtual badges were awarded to students for reaching a specific milestone (see Appendix C Figure 4). There are three badges; The bronze badge (starter): which the students get when they just start the lesson, the silver badge (Achiever badge) is given when the student completes at least two lessons and two game quests, and the gold badge (finisher) is given when the students completed all lessons and attempted all the games [88].	Engagement, performance [80], [76]
Quest	We incorporated interactive quests as part of the system using the Wordwall maze chase game. The goal was to encourage participants to use their HTML knowledge in practical situations. (See Appendix C Figure 3). Successful completion of the challenges contributed to the students' overall progress [60], [76].	Performance [30]
Leaderboard	A dynamic leaderboard displayed the rankings of the participants based on their position, attempts and time to finish the game [66] [76] (see Appendix C Figure 4).	Motivation [76], [66]

an open-ended question was included in the questionnaire to understand what the students think of the systems.

The student's performance in learning the subject using the two systems was measured using pre-test and post-test. Both groups took tests with the same questions, consisting of 15 multiple choice and true or false questions. The questions

were created upon validating the content and reviewed by experts to further validate the questions and related answers. The data that was later analyzed was collected from students who were asked to complete the pre-test to test their level of knowledge on the web development subject. After completing the course with the system, the students took the post-test to see their improvement and performance after using the system. A sample list of the test questions is seen in Appendix B.

The data collected was analyzed using the Statistical Package for Social Sciences (SPSS) software. Quantitative methods such as open-ended questions, and pre and post-test were used to statistically analyze and compare results, and to gather insights and determine the effectiveness of the game-based learning and gamification approach in improving students' motivation, engagement, and performance. Levene's test of homogeneity of variances [122] and the Reliability Statistics test [123] were also carried out to test assumptions and to measure the test consistency. The significant level used for testing each hypothesis for the three research questions was 0.05 with a 95% confidence interval. Overall, these analysis methods helped to provide a comprehensive understanding of the data.

In addition, a post hoc analysis was conducted using G*Power 3.1.9.4. The analysis was based on the actual sample sizes (Control: $n = 29$, Experimental: $n = 24$) and effect sizes calculated from the observed results. For engagement, the statistical power was 0.112 (Cohen's $d = 0.12$). Pre-motivation showed a power of 0.176 (Cohen's $d = 0.20$), while post-motivation had a power of 0.142 (Cohen's $d = 0.16$). These low power values suggest limited ability to detect small effects on engagement and motivation. On the other hand, the performance outcomes were analyzed using paired samples t-tests, and the effect sizes observed were significantly large. The experimental group's pre-post comparison showed Cohen's $d = -2.28$ with a power of 1.000, and the control group's pre-post comparison had Cohen's $d = -0.94$ with a power of 0.998. These results confirm that the sample size had very high power to detect performance improvements.

V. RESULT

A. MEASURING STUDENTS' ENGAGEMENT

The result of the observed engagement on the application of gamified system has been evaluated between the experimental and control group. Table 3 shows that there was no statistical difference found between that control and experimental groups' post-engagement. The Cronbach alpha test was conducted to confirm the internal consistency reliability of the engagement questionnaire. The initial analysis with 5 items (E1, E2, E3, E4), had 0.665 Cronbach alpha. After scaling the items and removing item E1 and E2, the test yielded a Cronbach's α of 0.843 indicating good reliability. A Shapiro-Wilk test for assumption of normality was performed to check if the data follows a normal distribution. This

TABLE 3. Mann-Whitney U test result to compare the engagement of control group and experimental group.

Groups	N	Mean Rank	Median	U	Z	p
Control	29	26.05	4.0	320.5	-.497	.619
Experimental	24	28.15	4.16			

Note. * $p < .05$.

TABLE 4. Control and experimental group's means and standard deviations for the pre and post-motivation.

Group		N	Mean	SD	Median
Control	Pre-motivation	29	3.70	.99	3.66
	Post-motivation	29	4.00	.61	4.00
Experimental	Pre-motivation	24	3.91	.53	4.00
	Post-motivation	24	4.06	.63	4.25

analysis indicated a significant difference between control group ($p = 0.020$) and experimental group ($p = 0.009$), suggesting the data is not normally distributed.

To evaluate the engagement difference between the control group and experimental group, A Mann-Whitney U test was performed. The results indicated no significant difference in the engagement of control group and experimental group, $U = 320.5$, $z = -.497$, $p = .619$, $r = .06$. Based on the result, Hypothesis 1 (H1) was not confirmed.

B. MEASURING STUDENTS' MOTIVATION

Cronbach alpha test was conducted to confirm the internal consistency reliability of the engagement questionnaire. The test yielded a Cronbach's α of 0.835 for pre-motivation and 0.759 for post-motivation indicating good reliability.

The pre-motivation and post-motivation measuring the motivation of the groups were not normally distributed, therefore the non-parametric Wilcoxon Signed-Rank test was applied within groups and, Mann-Whitney U test was applied between groups. Table 4 provides the means and standard deviations for the pre and post-motivation of the control and experimental groups.

There was no statistically significant difference in the pre-motivation and post-motivation for the control group, $z = -1.101$, $p = .271$, $r = .20$ and the experimental group $z = -.928$, $p = .353$, $r = .18$. This test revealed that the students' motivation did not significantly improve after using the gamified system.

To compare the pre-motivation and post-motivation between the control and experiment groups the Mann-Whitney U test was carried out. The analysis from Table 5 indicates that the pre-motivation of the control group and experimental group was not statistically significant, $U = 307$, $z = -.743$, $p = .457$, $r = .10$. Similarly, there was no statistically significant difference in the control group and experimental group's post-motivation, $U = 315.5$, $z = -.587$, $p = .557$, $r = .08$. Based on the result, Hypothesis 2 (H2) was not confirmed.

TABLE 5. Mann-Whitney U test result to compare the motivation of control group and experimental group.

Groups	<i>N</i>	Mean Rank	<i>U</i>	<i>Z</i>	<i>p</i>
Pre-motivation					
Control	29	26.05	307	-.743	.457
Experimental	24	28.15			
Post-motivation					
Control	29	25.88	315.5	-.587	.557
Experimental	24	28.35			

Note. * $p < .05$.

TABLE 6. Means and standard deviations for the pre-test and post-test based of the control and experimental groups.

Group		N	Mean	SD
Control	Pre-test	29	53.72	16.54
	Post-test	29	70.97	16.44
Experimental	Pre-test	24	46.33	18.05
	Post-test	24	82.38	13.52

C. MEASURING STUDENTS' PERFORMANCE

The pre-test and post-test measuring the performance of the students were normally distributed, therefore the parametric Paired Samples t-test was test was applied within groups and, independent t-test was applied between groups. Table 6 shows the means and standard deviations for the pre and post-test of the control and experimental groups. To determine if the mean score of the pre-test and post-test were different within the control and experimental group paired samples t-test was performed.

There was a statistically significant difference in the pre-test and post-test for the control group, $t(28) = -5.08$, $p = .000$ and the experimental group, $t(23) = -11.18$, $p = .000$. This test revealed that the students' performance significantly improved after the interventions. To compare the pre-test and post-post between the control and experiment groups the independent t-test was carried out. The analysis indicates that the pre-test of the control group and experimental group was not statistically significant, $t(51) = 1.55$, $p = .126$. This result indicates that the students were on the same level before the intervention. However, there was a higher statistically significant difference in the post test-test scores of the students in the experimental group as compared to the students in the control group, $t(51) = -2.72$, $p = .009$. Based on the result, Hypothesis 3 (H3) was confirmed.

D. SUMMARY OF STUDENTS' GAMEPLAYS

Gamified sample profile information on earned progress points (see Appendix C Figure 4) revealed that all the students in the experimental group earned points, badges, and levelled up, suggesting that they were actively engaged with the gamified system. The gameplay consisted of three games as shown in Table 7. Based on the results of the game plays, a considerable number of students played the games included

in the course, indicating that they were motivated to participate in the learning activities.

The result shows that students had high scores and completed the gameplay fast, which suggests that the gamified system was successful in promoting learning and performance. Most of the students achieved near-perfect scores and completed the gameplay quickly, suggesting that they had a good understanding of the material and could apply their knowledge effectively. However, from the gameplay results the number of students who played the game decreased towards the end, with 24 students out of 30 completing the final game. The drop-off rates were 13.3% in Game 1 and improved slightly to 10% in Game 2, suggesting a re-engagement rate of 3.3%. However, in Game 3, it increased to 20%. This result makes a total drop-off of 6.7% from Game 1 to Game 3 and 10% from Game 2 to Game 3. These drop-off rates suggest possible motivational fatigue or novelty effects.

E. QUALITATIVE ANALYSIS ON THE USE OF THE GAMIFIED SYSTEM

The students were given the opportunity to provide additional comments about the gamified system and their learning experience using an open-ended question. From the experimental group's responses, a few interesting points were noted, leading to the identification of two main themes.

The first theme (General feedback on the system), revolved around students expressing varied feedback and satisfaction with the gamified system. Some students enjoyed the experience and found it helpful in learning the course. Some even expressed a desire to continue using the system for other subjects. However, for some students the system was not efficient enough. Noteworthy responses made by the students are as follows:

"It was fun."

"I find games are a great way to engage students to learn and make them have fun at the same time. It can make some knowledge more memorable as they will associate it with a fun memory, or something they found striking in the game."

"It was engaging."

"I think the 'gaming' part wasn't really convenient to the user." "It feels like I'm solving a quiz that takes a long time to answer, rather than playing a game. But it surely helps you learn things a few or more."

"I like the system, using the system helped me understand the lesson."

"I think the game was not efficient enough."

"Learning web development through games is making the learning process easier and improving the motivation & desire to learn more."

"It was a great experience."

The second theme (Impact on learning outcomes) students reported that the system was good for new developers, and they had learned and understood the subject matter better through the gamified system experience. Noteworthy responses made by the students are as follows:

TABLE 7. Gameplays summary of the experimental group.

Gameplays	Game information						
	No. of Students	Submissions	Average Score	Top Score	Fastest Time (sec)	Participation Rate (%)	Drop-off Rate
1	30	26	3.9/4	4/4	18.6	86.7%	13.3%
2	30	27	4.5/5	5/5	19	90.0%	10%
3	30	24	9.3/10	10/10	20.7	80.0%	20%

“Playing the games helped me to learn the HTML very well.”

“I would like to learn my subjects with this gaming-learning method, thanks for the experience.”

“I think for new developers this is a good way to learn web development.”

“After taking the course, I can say it helped me understand this subject. Thank you.”

F. COMBINING QUANTITATIVE AND QUALITATIVE DATA

The quantitative and qualitative results are reciprocal with complementary insights into the impact of the gamified system on students' engagement, motivation, and performance. The students provide varied feedback about their experiences with the gamified system, including aspects of efficiency, engagement, motivation and learning outcomes.

Despite the quantitative results for engagement indicated that experimental group (Median = 4.16, $n = 24$) had slightly higher median score compared to the control group (Median = 4.0, $n = 29$), and that the experimental group's (Median = 4.25, $n = 24$) motivation score was also slightly higher than that of the control group (Median = 4.00, $n = 29$), these differences were not statistically significant enough. This lack of difference can be explained in the qualitative responses of the students.

Some students found the system fun, engaging, and helpful for understanding the subject, while others stated it was more like a quiz rather than a game. According to these mixed responses, the gamified system was interpreted differently by individual students depending on what they expected it to look like. For some students, the system provided motivating and engaging challenges, but for others, the format felt quiz-like, which may have reduced their motivation or sustained engagement.

On the other hand, the performance result revealed a clearer understanding of our findings. The experimental group showed significantly higher improvement in post-test scores compared to the control group. This is strongly supported by qualitative responses in which students reported that the gamified system helped them understand the subject better and reinforced their learning. These outcomes suggest that, even if engagement and motivation levels varied, the system was successful in fostering competence, focus, and mastery of the subject as a result of the gamified system.

VI. DISCUSSION AND CONCLUSION

The purpose of this study was to investigate the impact of a gamified system on engagement, motivation, and academic performance among higher education students with novice or beginner level education. Game-based learning and gamification elements were both integrated to create an interactive learning experience, with SDT and Flow Theory informing the psychological and motivational goals of the system. While our finding reported significant improvement in performance, the findings on engagement and motivation were more mixed.

Quantitative results showed a significant improvement in students' performance in the experimental group compared to the control group (H3). However, no significant difference was found in students' engagement and motivation between the experimental and control groups. This suggests that the system worked well for some students, but not enough to create a strong effect to show a significant improvement in their engagement and motivation. These mixed findings which are consistent with previous results seen in other gamified education studies and points to similar influencing factors such as nature of course content, game element design, individual differences, depth of intervention, technological recourse that can contribute to the overserved outcomes game [70], [124], [125], [128].

According to Hypothesis 1, we found that using the gamified system to learn web development did not improve the students' engagement in contrast to the non-gamified system. This result is similar to previous studies which showed that the integration of gamified learning can yield varied engagement results depending on the learning context [57], [126], [127].

Similarly, the difference in motivation was not statistically significant, meaning that the evidence of increased motivation was not strong enough to confirm H2.

In contrast to the engagement and motivation results, integrating game-based learning and gamification showed that the experimental group performed better than the control group. The game elements that were used in these study such as levels, points, badges, challenges, and leaderboards provide insights for the reason for the higher performance in the experimental group than the control group as evidenced by the earned points, badges levelled up and about 27 students attempted the game plays, as evidenced by fast completion times and high scores highlighting their active engagement and performance. In addition to this, the analysis of the

qualitative results showed that students were able to learn the subject, with students noting, “Playing the games helped me to learn the HTML very well.”, “After taking the course I can say it helped me understand this subject. Thank you.”

This study results are consistent with previous studies that show positive improvement in students’ performance after the use of gamified system [67], [69], [70].

A. THEORETICAL INTERPRETATION

The mixed findings of this study can be better understood through Self-Determination Theory (SDT) and Flow Theory, both of which highlight the importance of intrinsic motivation and sustained engagement in learning.

According to SDT, intrinsic motivation increases when learners feel autonomous, competent, and connected to others. While the game elements such as levels, badged, progress points, challenges, and leaderboard were chosen to address students’ psychological needs, the impact of the game mechanics depends not only on their presence but on how students perceived and interacted with them. Some students described the system as “fun” and “engaging”, indicating that the mechanics met these needs for some individuals. However, others described it “quiz-like” or “inefficient, which may have potentially limited their sense of autonomy or immersion. These responses show how individual learner differences, such as students’ prior game experience, digital tool familiarity, or learning preferences may shape how gamification is received [66].

Wordwall and LMS were used in this study to implement the gamified system. While the system provided immediate feedback and progression, it is important to acknowledge that the gameplays implemented with Wordwall may have limited the system’s effectiveness, particularly for students who were hoping for a more engaging or playful experience. This suggests that the game format might not have met all students’ expectations, which could have limited the system’s ability to sustain intrinsic engagement.

Furthermore, Flow Theory posits that engagement is highest when learners are in a state of flow, which is achieved when challenge and skill are balanced and accompanied by immediate feedback. Although the gamified system included challenges and progression systems, the decreasing student participation in the gameplay stages suggests a novelty effect [70] where initial interest declines over time once students become more familiar with the system, or motivational fatigue, where prolonged effortful behaviors can lead to decrease in willingness to continue a task [128]. Students’ perceptions of the system as “quiz-like” or “not convenient”, may have affected the students’ sense of progress or meaningful feedback required to sustain flow.

These interpretations reinforce that gamified learning outcomes depend not only on the inclusion of the game elements but on how they are experienced by learners, based on their preferences, expectations, and how well the designed system sustains engagement over time [129]. This aligns

with findings from a previous study [70], which found that while game elements can improve students’ engagement, they are not enough to sustain engagement as gamification can often suffer from novelty effect, with student interest dropping off unless the system is continuously supported by meaningful design elements. According to their results, sustained engagement was achieved when the gamified system included pedagogical factors such as coherent narrative, careful selection of learning tasks, clearer expectations, regular communication, and feedback.

B. CONTRIBUTIONS

The main contribution of this study is first; by integrating game-based learning and gamification approaches to learning web development. Our findings provide practical evidence that gamified systems, particularly for novice learners, when they are designed with clear goals, structured challenges, and mechanisms for feedback, can greatly enhance performance in web development education. The use of levels, points, badges, challenges, and leaderboards helped create an engaging experience that supported student learning outcomes, particularly in foundational topics such as HTML. Second, the study shows that while game mechanics can promote learning, they are not enough to significantly improve engagement and motivation. Students’ qualitative feedback made it clear that perceptions of the system’s design such as it feeling “quiz-like” or “inefficient” influenced how the gamified experience was received. This finding highlights the importance of aligning such systems with students’ expectations and preferences, to fully support flow, autonomy or sustained engagement over time. Third, this study contributes to theoretical understanding by applying SDT and Flow theory to design and evaluate the gamified system. These frameworks helped explain why some students remained engaged while others disengaged over time. Finally, this study contributes to the existing knowledge on game-based learning and gamification by providing further evidence of the effectiveness of this approach for improving web development education.

C. LIMITATIONS AND FUTURE WORK

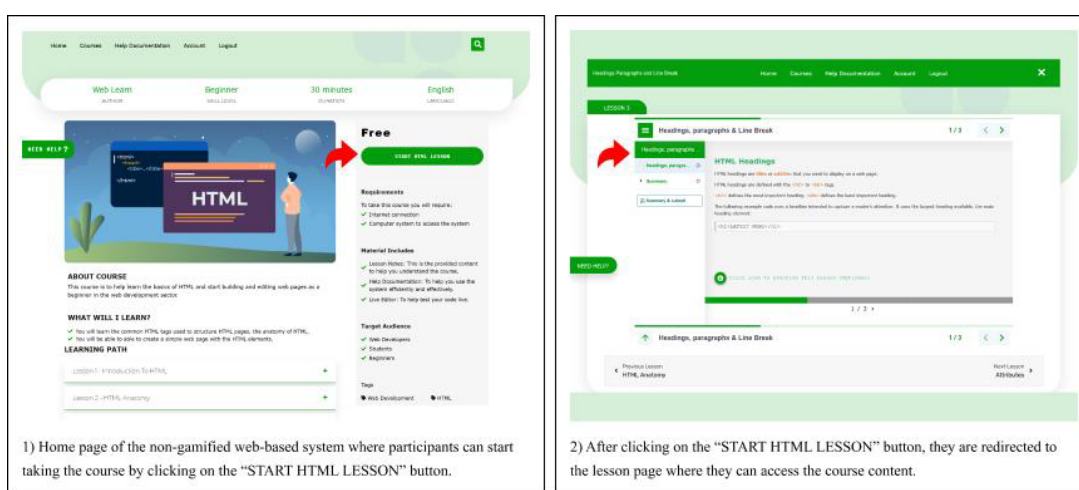
This study has some limitations: Although this study may have controlled for certain variables such as educational level, gender, and prior knowledge level of the subject, we acknowledge that students’ prior gaming experience or learning preferences were not measured. These variables could act as potential confounders, influencing how the students interacted with the gamified system. For instance, while Wordwall provided structured feedback and progression, some students perceived it as “quiz-like”, possibly due to a mismatch between the expected and experienced game format. However, it is important to also note that these students that felt the game was quiz-like that took time to answer, also acknowledged that the system helped in learning few or more things, indicating that they recognized the potential educational benefit of a gamified approach even though it

TABLE 8. Table of specifications.

Learning Objectives	Level of Objectives (Bloom’s Taxonomy)						Time spent (min)	% of time	Test Item	Test Item Types
	Lower Level		Higher Level							
	Remember, Understand, Identify		Apply, Analyze, Create, and Evaluate							
	Knowledge	Comprehension	Application	Analysis	Synthesis	Evaluation				
Lesson 1 (obj 1 – 2)	2	1					10	20%	3	2 Multiple Choice 1 True/False
Lesson 2 (obj 3 – 5)		1	2	1		1	15	30%	5	5 Multiple Choice
Lesson 3 (obj 6)	1			1	1		10	20%	3	3 Multiple Choice
Lesson 4 (obj 7 – 8)	1			1	1	1	15	30%	4	3 Multiple Choice 1 True/False
Total	4	2	2	3	2	2	50	100%	15	15

TABLE 9. Sample of the pre-test and post-test questions.

Questions	Options
1. What does HTML stand for?	A. Hyper Text Markup Language B. Hyperlinks and Text Markup C. Home Tool Markup
2. Locate the element that specifies the title for the HTML page.	A. <head> B. <title> C. <!DOCTYPE html>
3. Does HTML describe the structure of a web page?	A. True B. False
4. Consider the scenario “Adding an image element within an HTML <head> element”, the result will be: the image will be displayed on the web page. What is wrong with this result?	A. Nothing - the result is correct. B. Nothing is wrong with the result; this will only show the image at the top of the page. C. This will not display the image on the web page because content will only show when added within the <body> element.

**FIGURE 1.** An image of the non-gamified web-based system for the control group.

might have not met their preferences. This expectation may have limited their perceived autonomy or immersion, which affected their engagement and motivation. Future work could

address this limitation by testing gamified tools with students beforehand, incorporate learner feedback into tool selection, and explore more immersive or adaptive systems that can

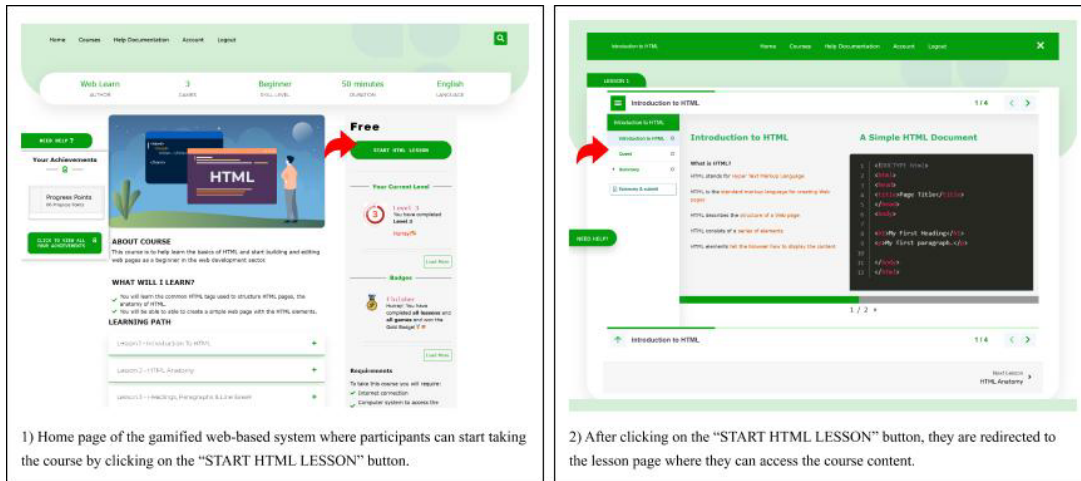


FIGURE 2. An image of the gamified web-based system for the experimental group.

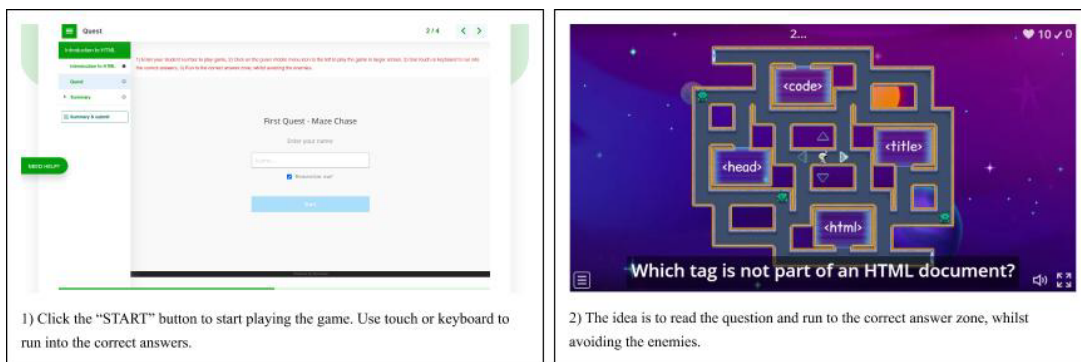


FIGURE 3. An example image of each quest in gamified web-based system.

better accommodate individual preferences and sustain motivation over time.

The quasi-experimental design, while appropriate for the education settings, introduced selection bias due to the absence of random assignments. Future work should consider collecting baseline data on important confounders such as students' prior gaming experience or learning preferences to statistically control their effects. For example, using pre-test scores as covariates in the analysis or applying techniques such as propensity score matching (PSM) to reduce the selection bias in quasi-experimental designs.

Post hoc power analysis revealed that the sample size may have been insufficient to detect small effects in engagement and motivation. The estimated power to detect small effects ($d < 0.20$) ranged between 0.11 and 0.17, which is far below the conventional threshold of 0.80 [130], [131]. This means that even if a true effect existed, our sample size might not have been enough to detect it. Therefore, the absence of significant results in these variables should be interpreted with caution, and future studies with larger samples are recommended to provide more conclusive evidence [130], [132].

The intervention was conducted for a short period and focused on introductory topics. As such, the long-term impact

of the gamified approach and its effectiveness for more advanced topics remains unexplored. Future studies should consider extended interventions that progressively align game difficulty with learners' skills, an important aspect of sustaining learners' flow.

Although some students remained engaged throughout the intervention, participation in the final gameplay decreased. We observed a 3.3% drop-off from Game 1 to Game 2 and a 10% drop-off from Game 2 to Game 3, suggesting a possible novelty effect or motivational fatigue, as identified in prior studies [70]. One possible explanation for this can be assumed to be the design of the gamified system itself, which incorporated traditional elements such as points, badges, levels, and leaderboards. While these elements are commonly linked to extrinsic motivation, they may not fully support the psychological needs of autonomy, relatedness, and intrinsic engagement described by SDT and Flow Theory. As a result, some students may have experienced disengagement over time when these needs were not sufficiently met. To address this, future studies could consider problem-based gameplay, narrative experiences, or collaborative challenges to provide more to create more immersive learning experiences. These approaches may not only reduce the novelty effect but may

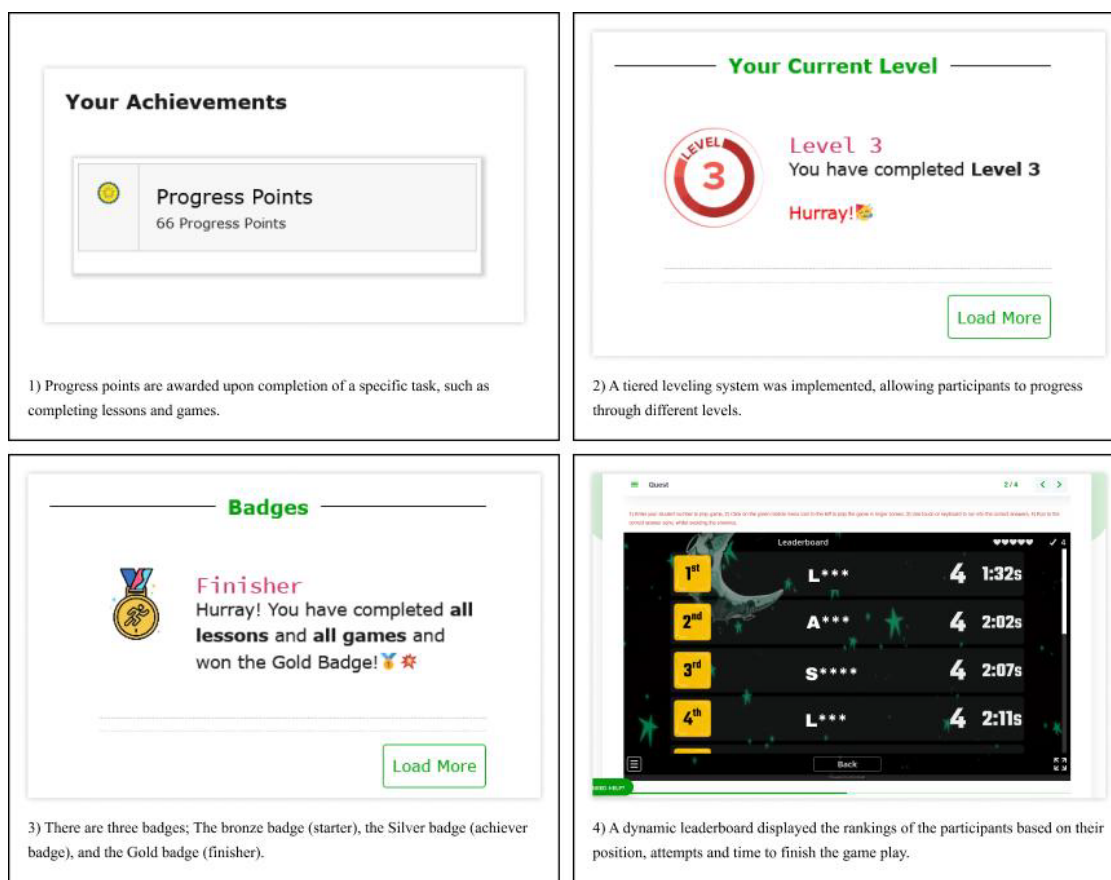


FIGURE 4. An image of the game elements considered in designing the gamified system.

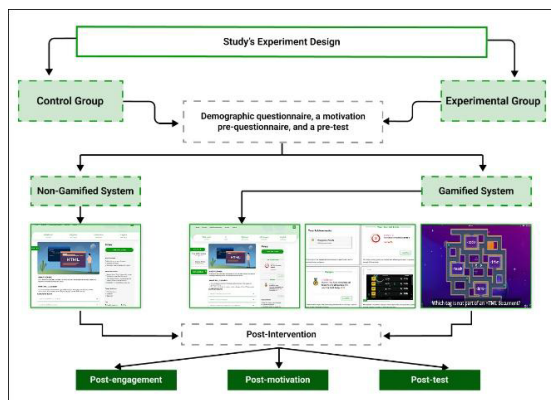


FIGURE 5. Experimental design of the study.

help sustain engagement over time and better align with the motivational needs of diverse learners.

In this study, to explore the impact of gamified experience, we used a combination of multiple game elements (points, badges, levels, leaderboards, and challenges) at the same time. This approach also limited the ability to identify the specific mechanics that contributed most to the observed effects. Therefore, to disentangle the specific contributions of each gamification component, future studies could employ sequential A/B testing or a factorial experimental design.

This study consisted of undergraduate engineering students from Turkey, which limits the generalization of our findings. Therefore, we recommend that future work replicates this study across diverse cultural settings and institutional and academic disciplines to validate and extend our findings.

Finally, future studies could explore research questions such as (1) How can adaptive difficulty gamification be used to sustain engagement as learners' web development skills develop over time? (2) How do social competitive mechanisms, such as peer leaderboards, affect learning results and motivation in web development education? (3) What is the impact of mobile-based gamification on students' engagement, accessibility, and performance in web development?

APPENDIX A TABLE OF SPECIFICATION

See Table 8.

APPENDIX B SAMPLE TEST QUESTIONS

See Table 9.

APPENDIX C SAMPLE USER INTERFACE AND GAMEPLAY

See Figures 1–4.

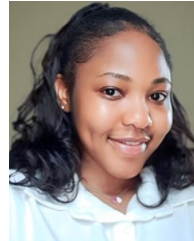
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